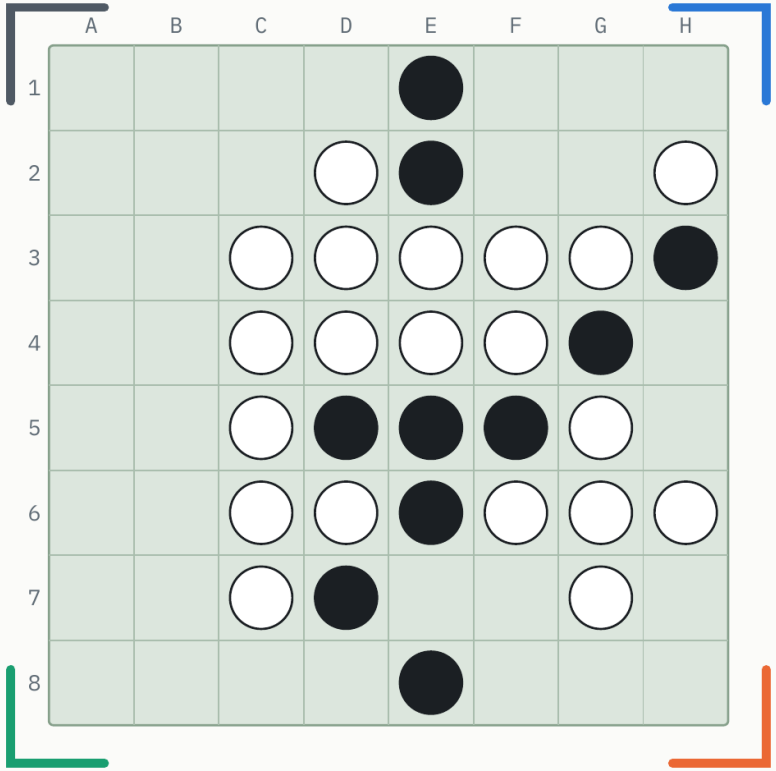
Do world-model diagnostics agree?

WM-Concord: four tests for a world model, run on the same models with the same ground truth

Venkatesh Nagarjuna · Siddardha Reddy · Nikhil Obuleni · Viharika Vemparala

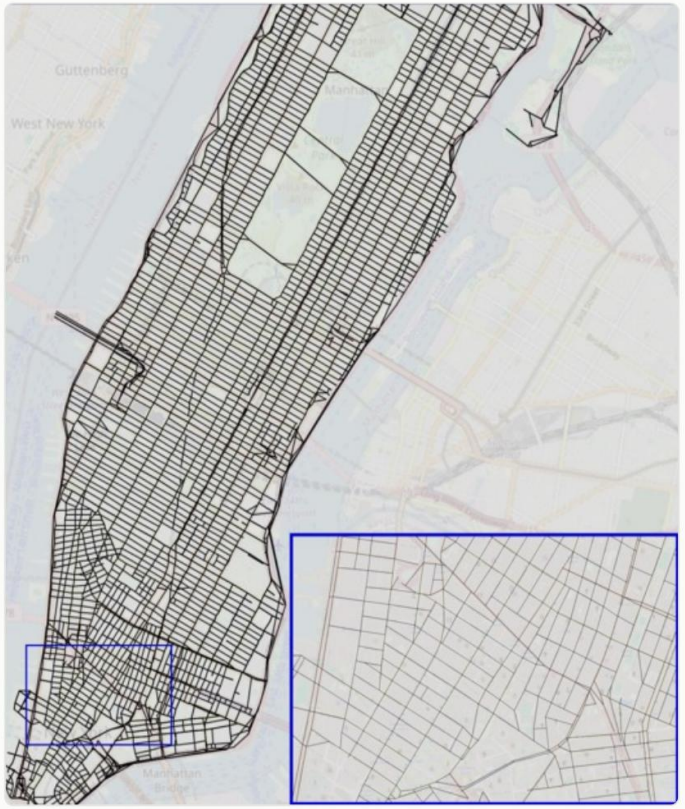
DATS 6501 DATA SCIENCE CAPSTONE · GROUP 6
THE GEORGE WASHINGTON UNIVERSITY · ADVISOR: DR. AMIR JAFARI
PRELIMINARY PRESENTATION · OCTOBER 20, 2026



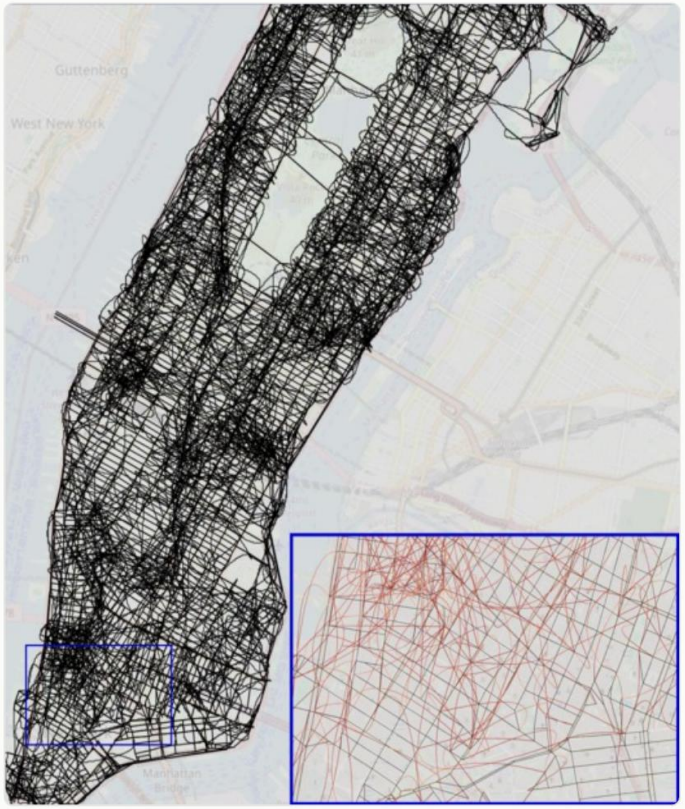
WHY THIS MATTERS

A model can follow the rules without knowing the world

Map implied by the true rules



Map implied by a trained transformer



Transformers trained on turn-by-turn NYC taxi routes:

- propose a **valid next turn** nearly 100% of the time
- let a probe recover the current intersection **> 90%** of the time
- yet imply a street map full of **streets that do not exist**
- and their routing **breaks when a detour is forced**

> Passing one test is not the same as having a coherent world model.

Images: Vafa et al. (2024), Fig. 3 (a) and (c), CC BY 4.0. Right: transformer trained on random walks; red edges in the inset are not in the real street graph.

THE GAP

Four families of tests, opposite verdicts, never run head to head

D1 Next-move validity

LOOKS AT

outputs, one position at a time

USED BY

almost every paper (Li et al. 2023)

OTHELLO VERDICT

high for both Othello-GPTs

D2 State equivalence

LOOKS AT

outputs across pairs of histories

FROM

Vafa et al. (2024)

OTHELLO VERDICT

synthetic model coherent, championship model not

D3 Inductive-bias probe

LOOKS AT

how the model adapts to a new task

FROM

Vafa et al. (2025)

OTHELLO VERDICT

weak bias toward the true board

D4 Representation probe

LOOKS AT

the board inside the activations

FROM

Li et al. (2023); Nanda et al. (2023)

OTHELLO VERDICT

board decodable, and edits change moves

> Each paper used its own models, data and code, so we cannot tell whether the tests disagree because they measure different things or because they were applied differently.

Related: state tracking in chess language models (Toshniwal et al. 2022); probing fine-tuned LLMs on Othello (Yuan and Søgaard 2025).

THE QUESTION

We treat “has a world model” as a construct with four competing measures

Do the diagnostics agree, and which one predicts when a model breaks?

	HYPOTHESIS	KIND OF VALIDITY
H1	Do the four diagnostics rank our models **in the same order**?	convergent
H2	Where they disagree, is it **systematic** (architecture, size, data) or noise?	structure of disagreement
H3	Which diagnostic best **predicts failure on shifted tasks**, beyond the legal-move rate?	predictive
H4	Do H1 to H3 hold on a **held-out domain**?	generalization

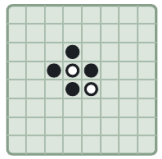
> We propose no new diagnostic. Agreement and disagreement are both reportable results.

Measurement framing: constructs, operationalizations, convergent and predictive validity (Jacobs and Wallach 2021); benchmarks that claim more than they measure (Raji et al. 2021).

DESIGN

One zoo, four instruments, one frozen yardstick

1 · BUILD



Othello rules engine

true board after every move
legal next moves
the only copy of the rules

game data (moves)

Model zoo

81 models per domain



3 architectures
× 3 sizes × 3 datasets
× 3 seeds

trained on moves only
never sees a board

2 · MEASURE

1 Next-move validity
outputs, one position at a time

2 State equivalence
outputs across pairs of histories

3 Inductive-bias probe
how fast it learns a board question

4 Representation probe
the board inside the activations

Fragility suite v1
same rules, shifted situations · frozen first

fragility

3 · COMPARE

Master table

one row per model, one score per test



higher always =
more world-model-like

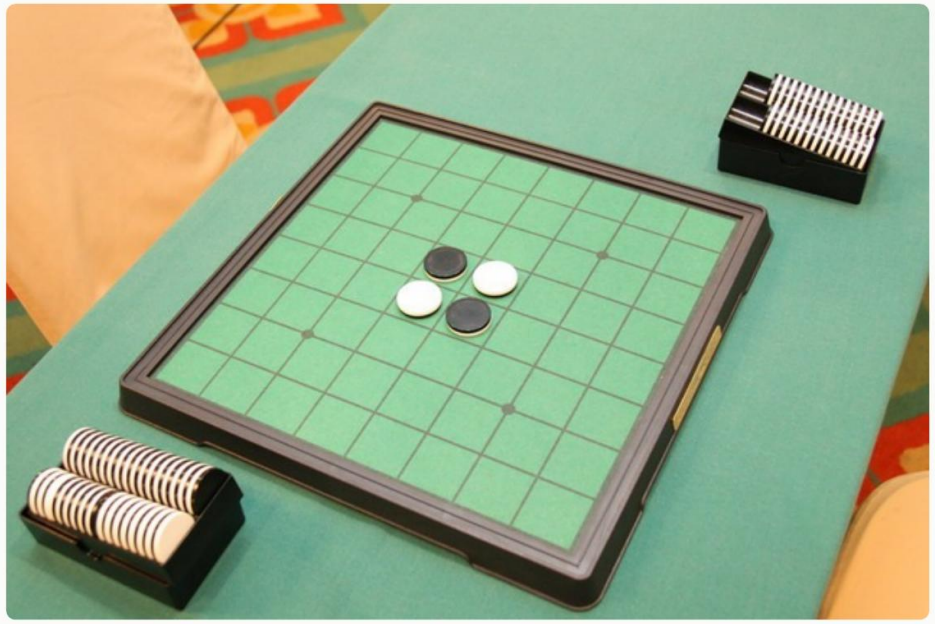
H1 Do the four rankings agree?
Kendall τ, also with competence partialled out

H2 Is disagreement systematic?
vs architecture, size, data; seed noise floor

H3 Which score predicts fragility?
beyond the legal-move rate (D1)

H4 · run all of it again on a held-out domain (the interpreter)

Othello: a world whose true state we always know



Competition board at the start position. Photo: Zen ar218, Wikimedia Commons, public domain.

- **State** = the 8 × 8 board + whose turn it is: a finite automaton our rules engine replays exactly
- **Models see only move tokens:** 60 squares + padding, never a board

Synthetic
random legal games, from the Othello-GPT corpus (Li et al. 2023)

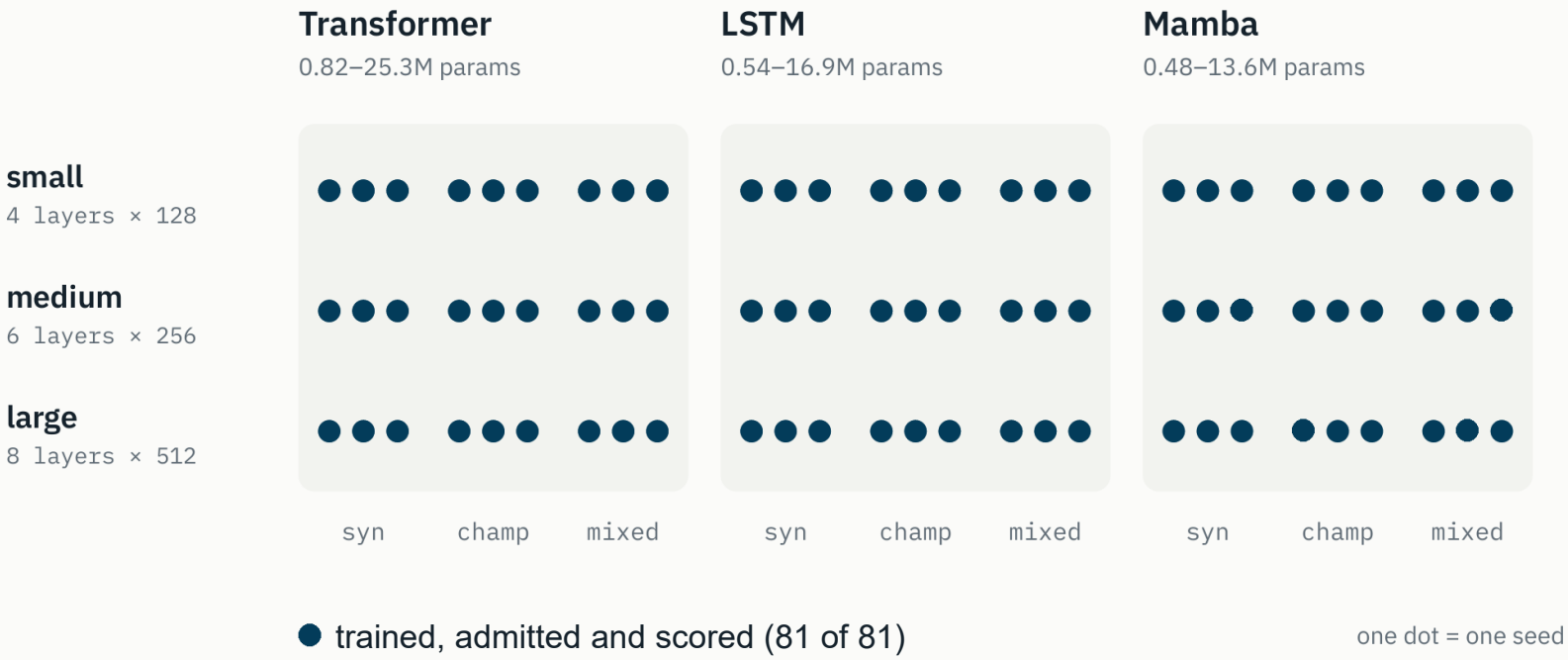
Championship
WTHOR archive: 137,548 human tournament games, 1977–2025

Mixed
both pooled; validation and test sets are 50 / 50

Splits: train → training · val → fitting probes and adaptation · test → final scores

MODEL ZOO

81 small sequence models per domain, deliberately varied



- **Architectures:** GPT-style transformer, stacked LSTM, Mamba
- **Same budget for all:** 5,000 steps × 256 games, AdamW, lr 3e-4
- **0.48 M to 25.3 M parameters**
- **Admission:** validation legal-move rate ≥ 0.50 (untrained ≈ 0.15)
- **3 domains × 81 = 243 models;** Othello first

Architectures: Vaswani et al. (2017); Hochreiter and Schmidhuber (1997); Gu and Dao (2023) (run with the pure-PyTorch mambapy build).

DIAGNOSTIC 1 · BEHAVIOR

Next-move validity: is the top prediction legal?

MOVES SO FAR

D3 C3 C4 → **model** (top-1 next move) → PREDICTION **E3**

THE CHECK

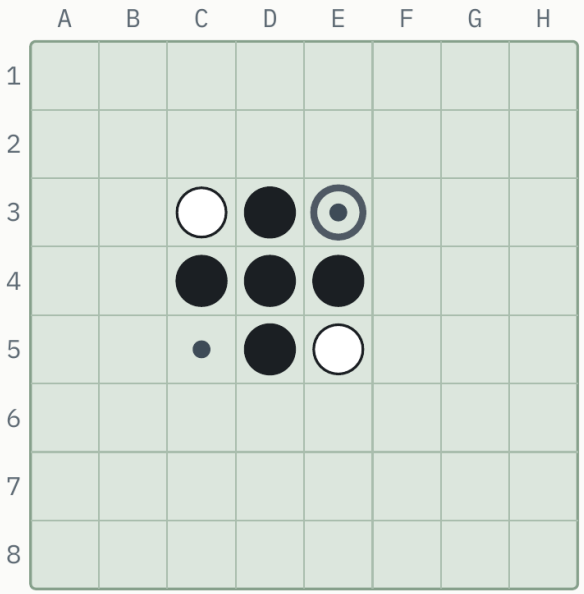
Rules engine replays the same moves:

`legal next moves = { C5, E3 }`

E3 ∈ legal set → valid (counts as 1)

C3 would be occupied → invalid (0)

Score = share of positions whose top move is legal. It does not need to match the move actually played (here C5): any legal move counts.



board after D3 C3 C4 · white to move

LOOKS AT

outputs at one position

SCORE

share of positions whose top move is legal

FITTED ON

nothing

DATA

1,000 test games of the model's own dataset

ANCHORS

synthetic **0.9996** · championship **0.896**

State equivalence: same board, same behavior

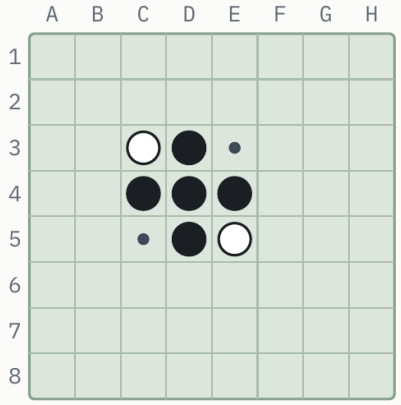
Compression same board, different order

HISTORY H

D3 C3 C4

HISTORY G

C4 C3 D3



one board · legal {C5, E3}

The model must accept the same moves after h and g.

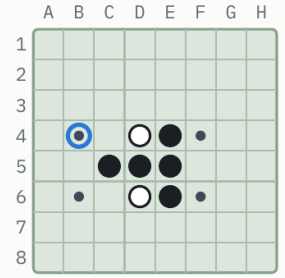
$$\text{error} = |A(h) \triangle A(g)| \,/\, |A(h) \cup A(g)|$$

A(·) = the model's top-k squares, k = number of legal moves

Distinction different boards, same length

HISTORY H

E6 D6 C5

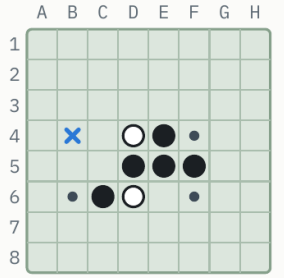


B4
legal

legal {B4, B6, F4, F6}

HISTORY G

F5 D6 C6



B4
not legal

legal {B6, F4, F6}

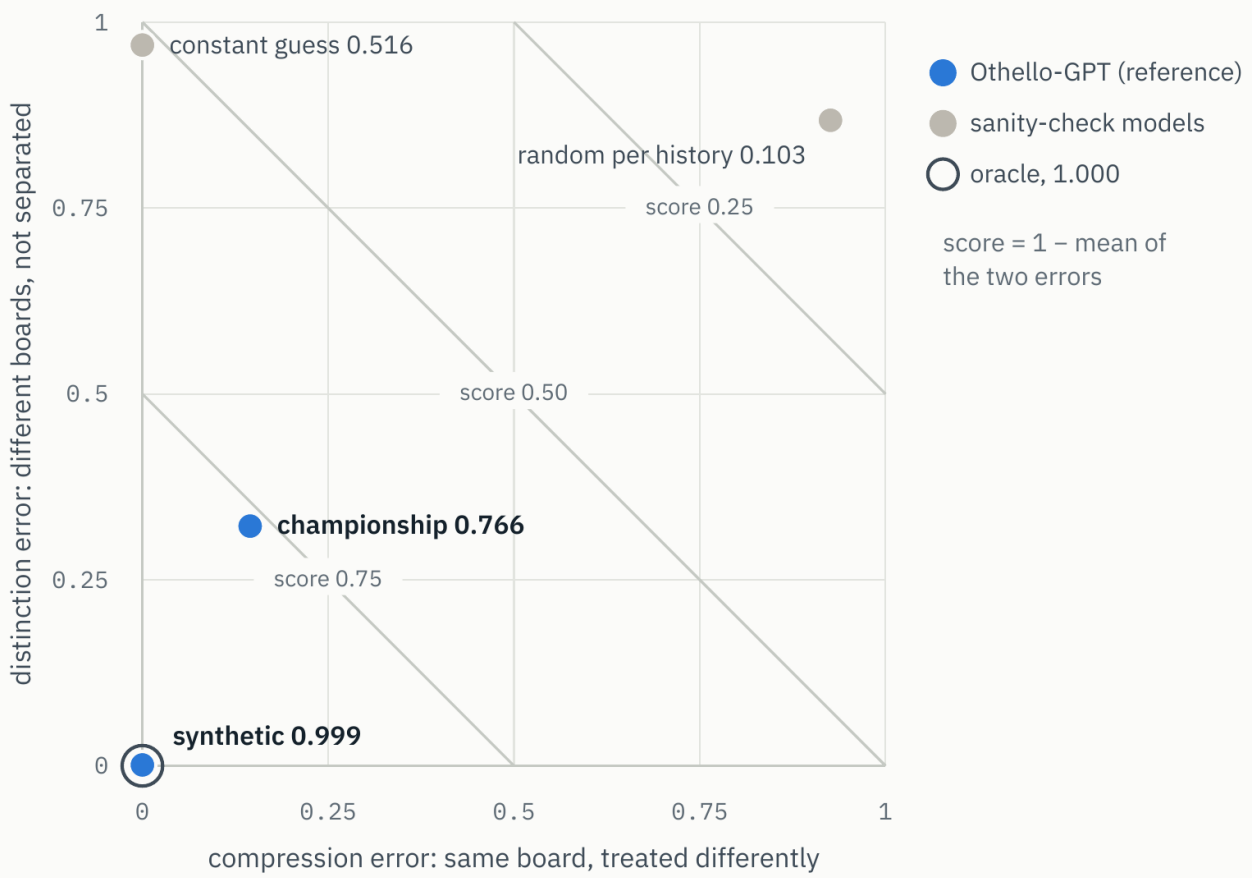
Boundary = moves legal after exactly one history: B4.

Accept B4 after h and reject it after g.

error = share of boundary moves the model gets wrong

Myhill–Nerode: two histories are the same state exactly when every continuation is equally allowed after both (Nerode 1958). Metrics after Vafa et al. (2024); examples replayed with our rules engine.

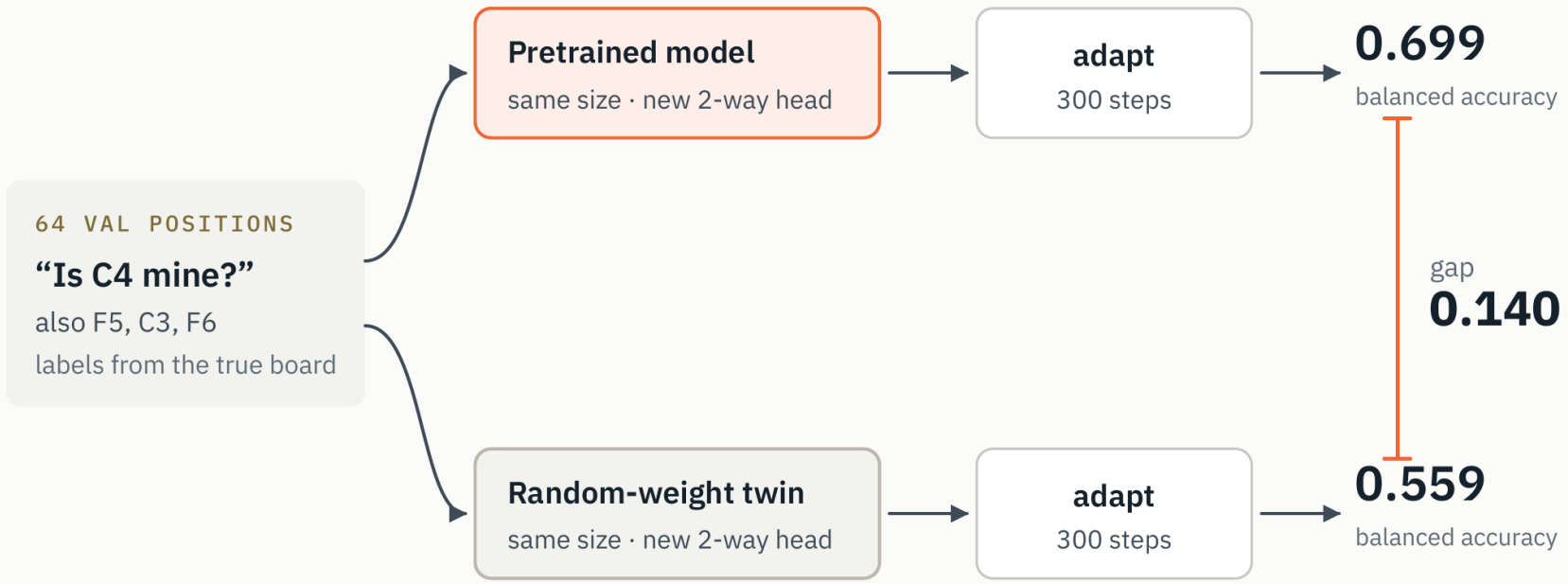
Two errors, because either one alone can be gamed



- **Score = 1 – mean of the compression and distinction errors**
- A **constant guess** never confuses two histories (compression 0) but separates nothing (distinction 0.97)
- Up to **2,000 pairs** of each kind → standard error ≤ 0.011; measured ≤ 0.005
- Anchors: synthetic **0.999**, championship **0.766**, the same order as Vafa et al. (2024)

DIAGNOSTIC 3 · ADAPTATION

Inductive-bias probe: does pretraining help learn a board question?



LOOKS AT
how the model adapts to a new task

SCORE
balanced-accuracy gap over a random-weight twin

FITTED ON
64 validation positions per task

WHY “MINE”, NOT “OCCUPIED”
occupancy can be read off the move list

ANCHORS
synthetic **0.140** · championship **0.059**

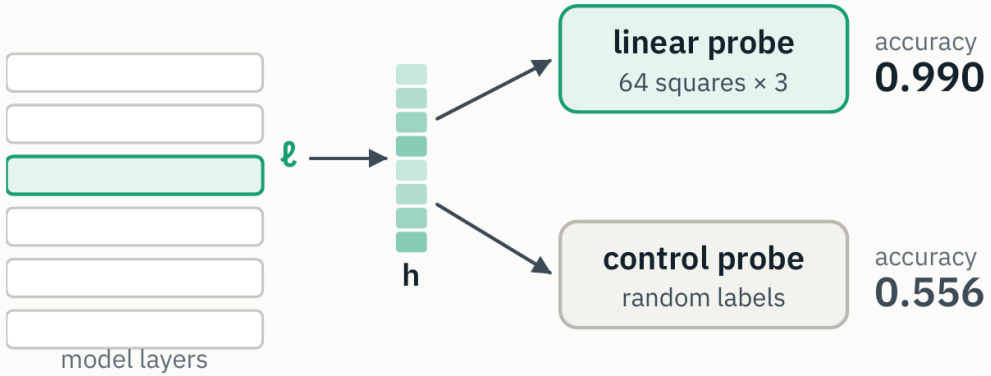
Othello-GPT synthetic, mean over 4 tasks × 3 seeds. Head lr 1e-3, body lr 1e-4, AdamW, batches of 16.

After Vafa et al. (2025).

DIAGNOSTIC 4 · INTERNALS

Representation probe: can the board be read out, and is it used?

1 · READ THE BOARD OUT



`selectivity = probe − control = 0.435`

layer chosen on validation games, scored on test games

Numbers: Othello-GPT synthetic, best layer 6 of 8.

2 · IS IT USED?

edit h: flip one disc
along the probe, middle → last layer

random edit, same size
the control for the edit

↓

do the predicted moves follow the edited board?

`causal score = probe − random edit`

on the moves whose legality flips

label: decodable and causal if selectivity ≥ 0.10 and causal score ≥ 0.20

LOOKS AT
activations after every layer

SCORE
selectivity of a linear probe at the best layer

FITTED ON
10,000 val games; layer chosen on 1,000 more

ALSO REPORTED
MLP probe · occupancy-only baseline · causal label

ANCHORS
synthetic **0.435** · championship **0.272**

Control task: Hewitt and Liang (2019). Causal edit after Li et al. (2023). Limits of probing: Belinkov (2022).

DEMO

Streamlit app: pick any model, watch each diagnostic

- **Model zoo:** registry and training record
- **D1 live:** a real game replayed move by move: the board, the legal moves, the model's pick
- **D2, D3, D4:** the saved result for that model, per layer or per task
- **Fragility:** drop per shift, v1 and v2
- **Concordance:** master table and rank agreement for all 81 models

```
streamlit run demo/app.py
```

Live demo

switch to the browser

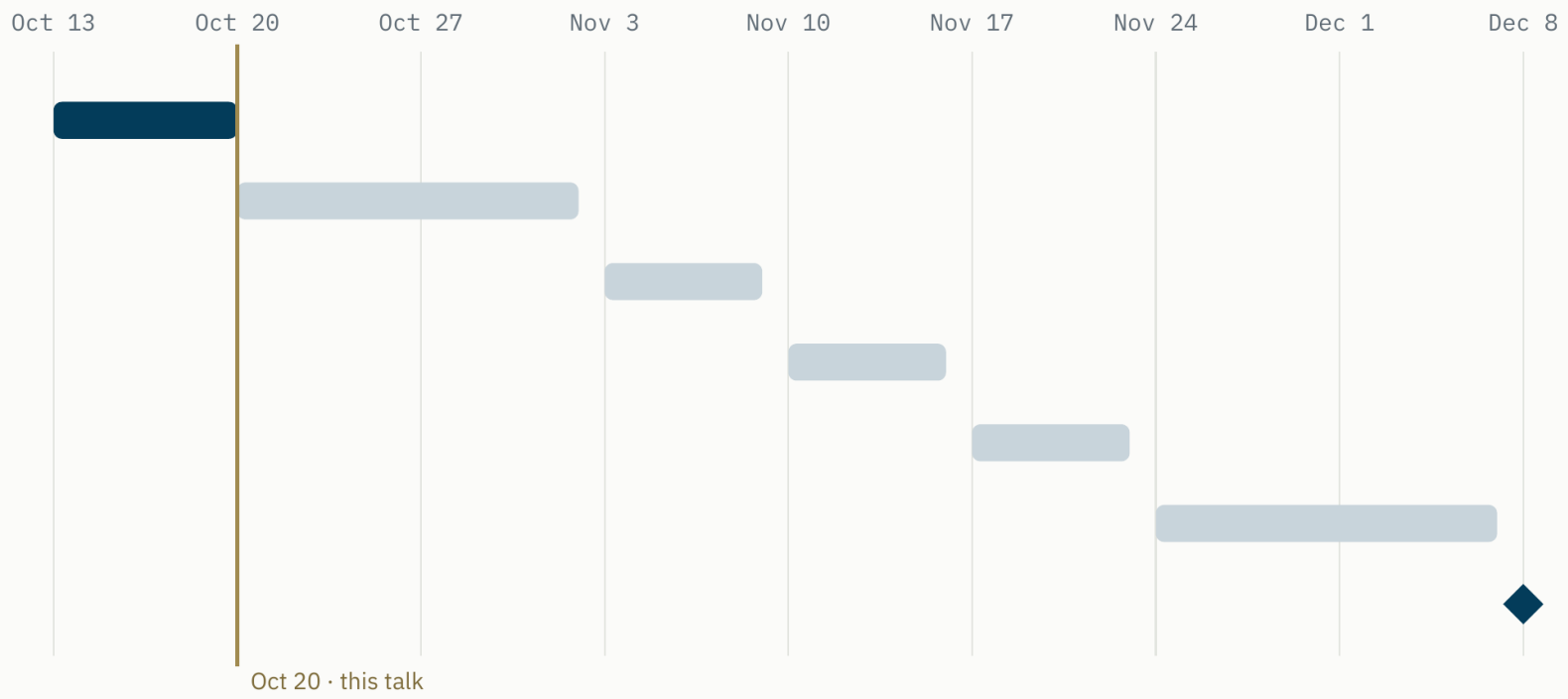
WHERE WE STAND

Status against the plan for today

DONE	Reproduction gate passed (Nanda et al. 2023, 5 of 6 within 1 point)
DONE	Fragility suites v1 and v2 frozen; preregistration with four amendments tagged
DONE	D1 to D4 implemented, tested, run on both reference Othello-GPTs
DONE	Othello zoo: 81 of 81 trained and admitted
DONE	D1 to D4 and both fragility suites on all 81 models: 486 results
DONE	H1, H2, H3 on Othello (preregistered), plus labelled exploratory checks
DONE	Literature review in the LaTeX report
IN PROGRESS	Demo on the real results; reply to Instructor Review 5
IN PROGRESS	Methods and Othello results in the report; checkpoints to S3
NEXT	Navigation domain and zoo; interpreter held out for H4

NEXT

Roadmap to December 8



Othello: zoo, D1–D4 on every model, H1 table

Navigation domain built; reproduce Vafa et al. 2024

Navigation zoo (81 models) and all four diagnostics

H1, H2 on two domains; interpreter frozen and trained, held out

H3; master table frozen

H4 on the interpreter; paper draft

Final presentation, paper, code release

TEAM

Who did what

Siddardha Reddy

LEAD ON

coordination · D4 probing and the reproduction gate · rules engine, data, interface · preregistration

Nikhil Obuleni

LEAD ON

zoo trainer and throughput · D3 inductive-bias probe · fragility suite · CI · demo app

Viharika Vemparala

LEAD ON

D1 next-move validity · report skeleton · data and splits write-up

Venkatesh Nagarjuna

LEAD ON

D2 review and preregistration sign-off · proposal and progress reports · this presentation

> Zoo training was split across our four instances, one slice of about 20 models each.

SUMMARY

Takeaways

1

Agreement is competence

The four diagnostics agree on which models are better (W 0.87), but with competence controlled they disagree (W 0.27). D1 is competence.

2

Disagreement is noise

Where they disagree, architecture, size and training data do not explain it (H2).

3

No reliable fragility predictor

D4 adds to D1 in the preregistered test, but the signal comes from training data, not from comparable models.

> **For anyone using these tools:** do not trust a single world model diagnostic, and always control for competence.

Next: navigation (reproducing Vafa et al. 2024), then the held out interpreter for H4; paper by December 8.

A readable board is not always a used board: 71 of 81 zoo models decode the board but do not use it.

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Thank you

Questions?

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BACKUP SLIDES FOLLOW

